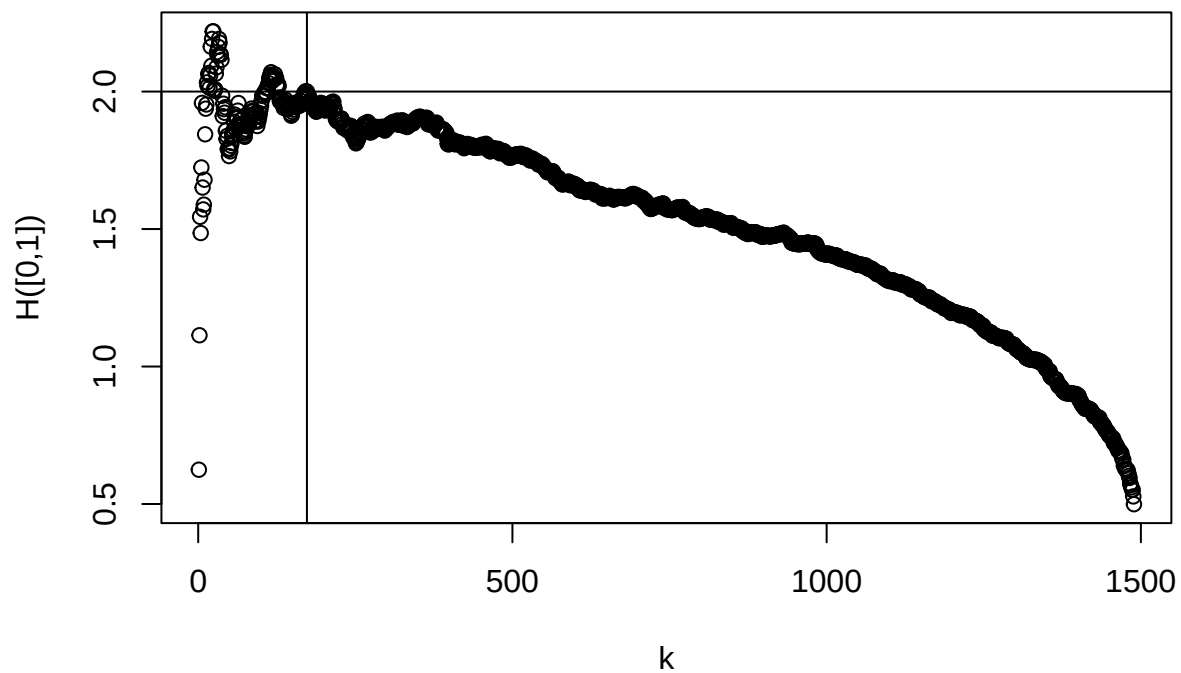


Supplement material: Model diagnostics

This notebook contains the different diagnostic plot for model goodness of fit

Threshold selection



graphical diagonotics of margins

BPOT margins

```
##      lower  estimate    upper
## 0.00000000 0.04667344 1.59181375
```

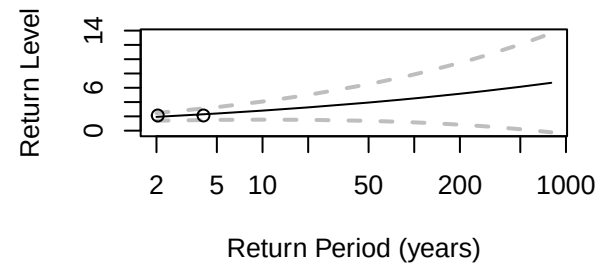
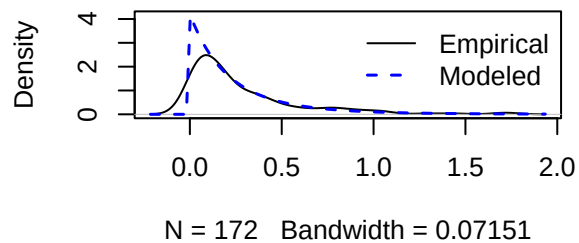
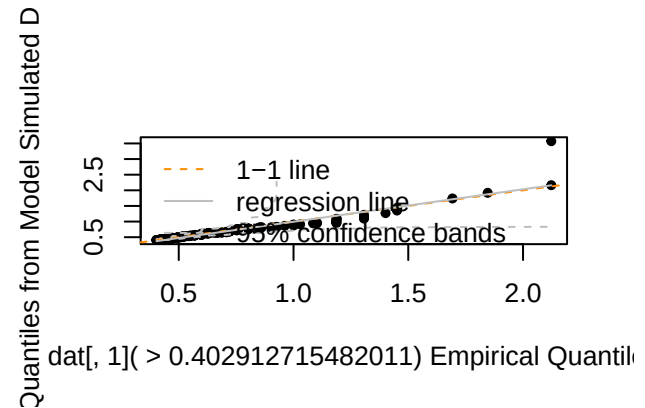
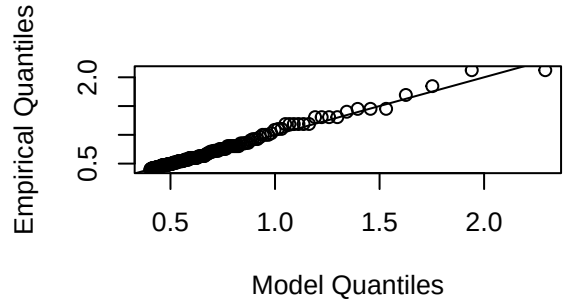
```
## [1] 0.00361683 0.52698278
```

```
## fevd(x = dat[, 1], threshold = thres[1], type = "GP")
##
```

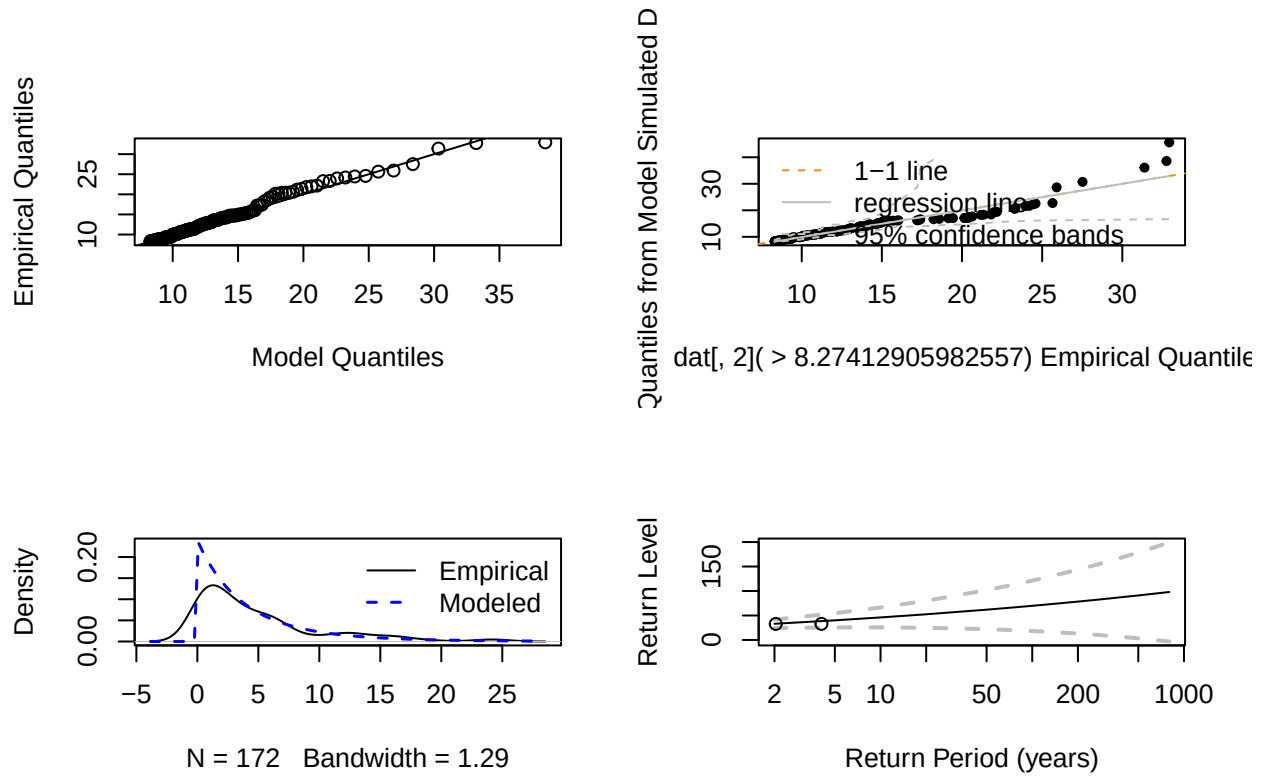
```
## [1] "Normal Approx."
##
##      95% lower CI  Estimate 95% upper CI
## scale  0.18035884 0.2377577   0.2951565
## shape -0.03351844 0.1575505   0.3486195

## fevd(x = dat[, 2], threshold = thres[2], type = "GP")
##
## [1] "Normal Approx."
##
##      95% lower CI  Estimate 95% upper CI
## scale  3.11316661 4.1753854   5.2376041
## shape -0.08145035 0.1252745   0.3319994
```

```
fevd(x = dat[, 1], threshold = thres[1], type = "GP")
```



```
fevd(x = dat[, 2], threshold = thres[2], type = "GP")
```

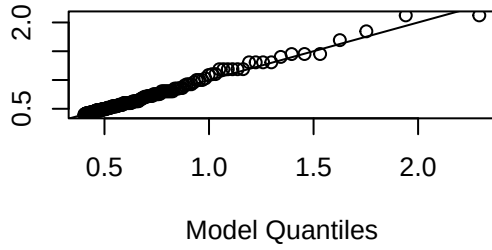


CBPOT margins

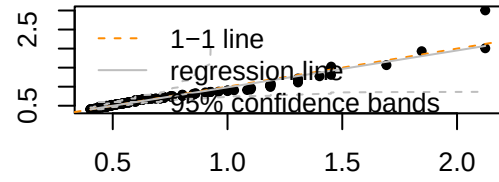
```
## [1] "CBPOT GP margin"
##
## [1] "Normal Approx."
##
##      95% lower CI  Estimate 95% upper CI
## scale  0.18035884 0.2377577  0.2951565
## shape -0.03351844 0.1575505  0.3486195
```

"CBPOT GP margin"

Empirical Quantiles

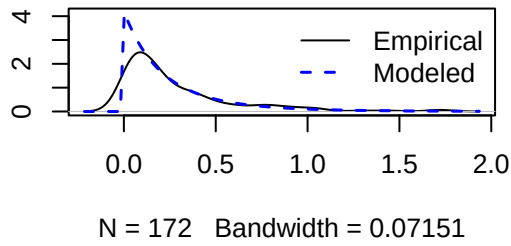


Quantiles from Model Simulated D

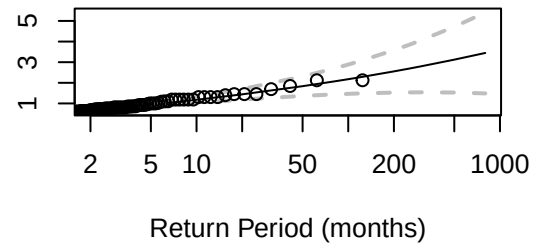


prox(> 0.402912715482011) Empirical Quantile:

Density

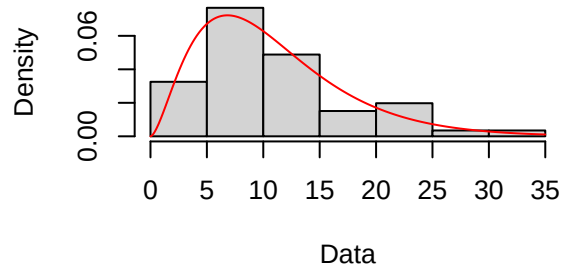


Return Level

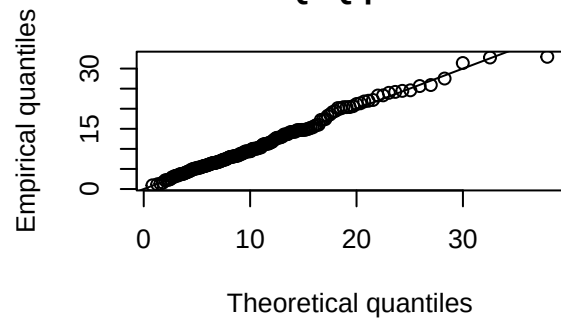


```
##          2.5 %    97.5 %
## shape 2.1483403 3.2198467
## rate  0.1926549 0.3009862
```

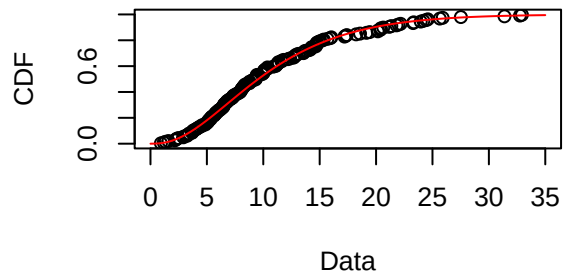
Empirical and theoretical dens.



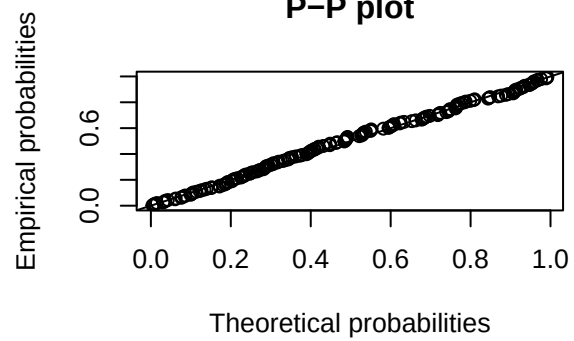
Q-Q plot



Empirical and theoretical CDFs

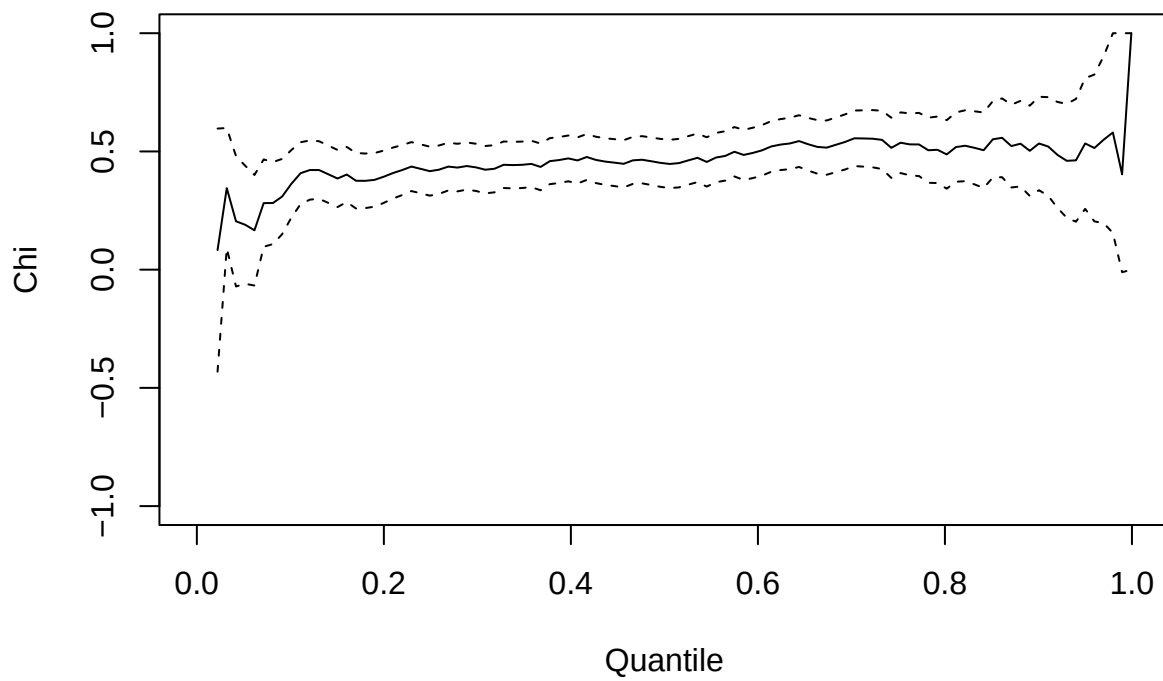


P-P plot

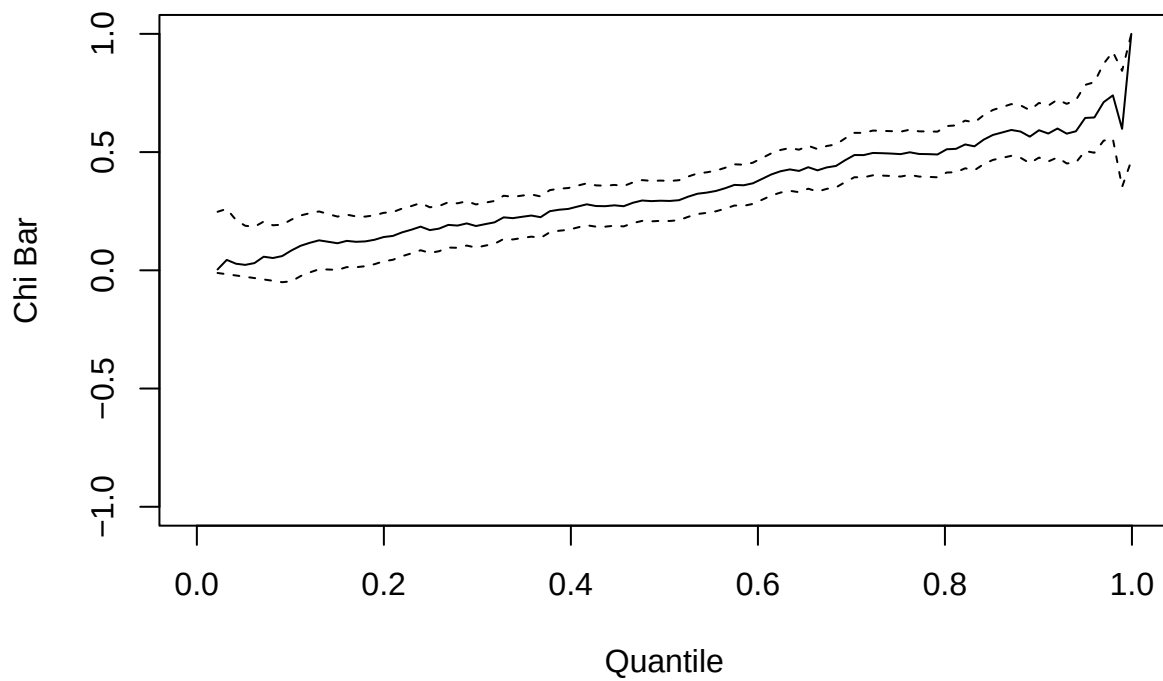


extremal dependence

Chi Plot



Chi Bar Plot



Copula GoF

```
##
## Parametric bootstrap-based goodness-of-fit test of Frank copula, dim. d
## = 2, with 'method'="Sn", 'estim.method'="mpl":
##
## data:  x
## statistic = 0.04865, parameter = 2.8393, p-value = 0.006494

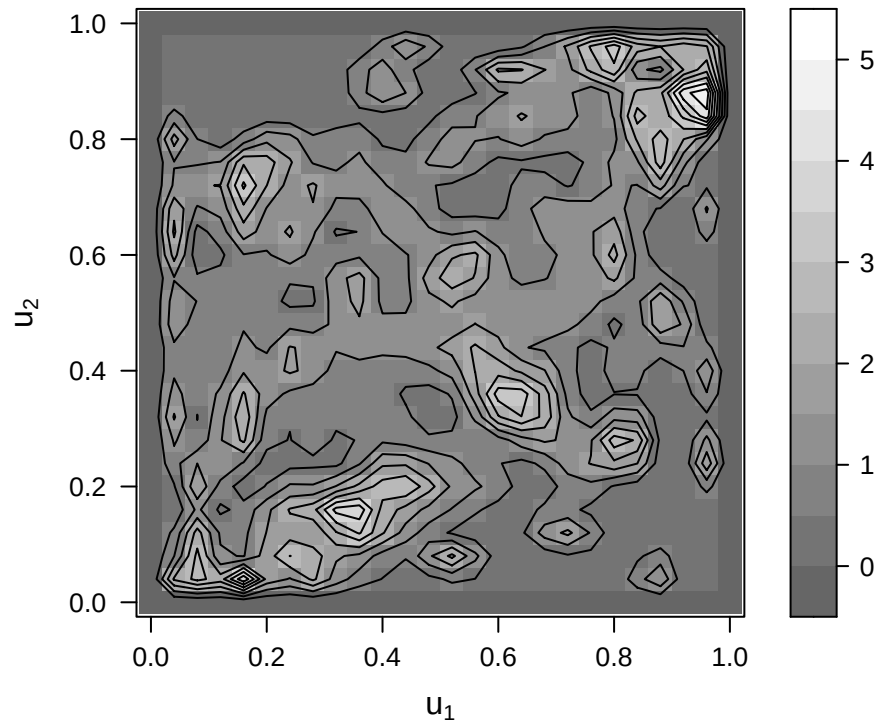
##
## Parametric bootstrap-based goodness-of-fit test of Clayton copula, dim.
## d = 2, with 'method'="Sn", 'estim.method'="mpl":
##
## data:  x
## statistic = 0.093035, parameter = 0.82545, p-value = 0.0004995

##
## Parametric bootstrap-based goodness-of-fit test of Normal copula, dim.
## d = 2, with 'method'="Sn", 'estim.method'="mpl":
##
## data:  x
## statistic = 0.039069, parameter = 0.47711, p-value = 0.03047

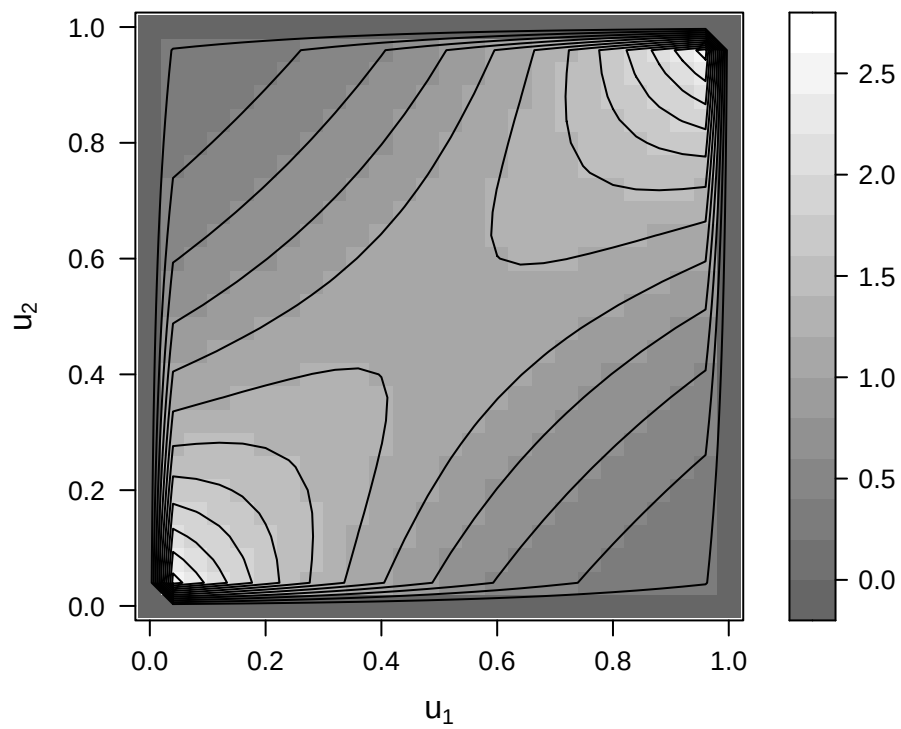
##
## Parametric bootstrap-based goodness-of-fit test of Gumbel copula, dim.
## d = 2, with 'method'="Sn", 'estim.method'="mpl":
##
## data:  x
## statistic = 0.021399, parameter = 1.4483, p-value = 0.3551
```

Copula plots

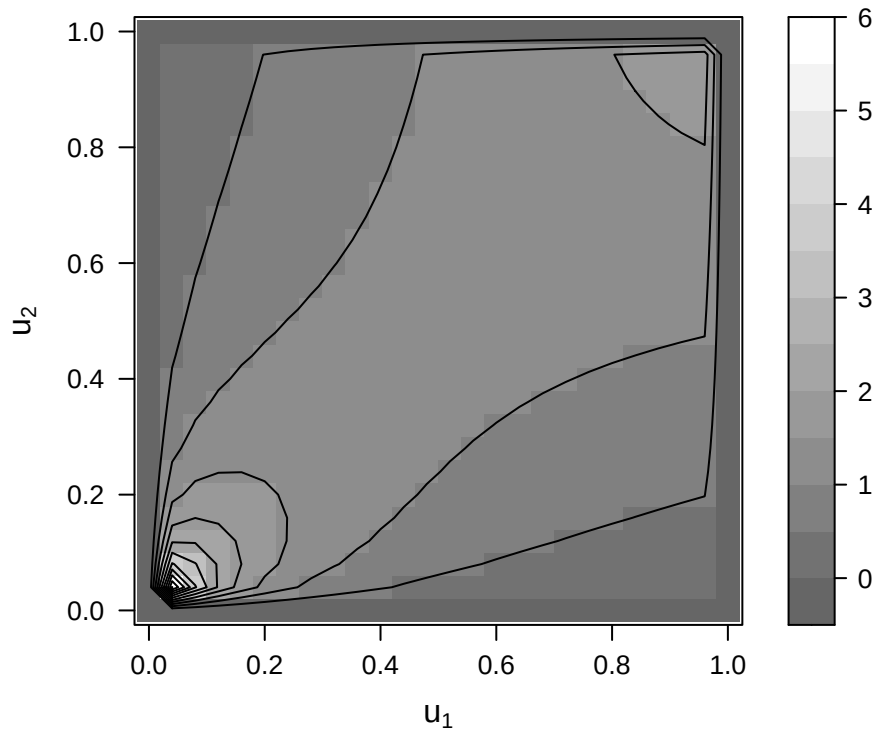
Empirical Copula



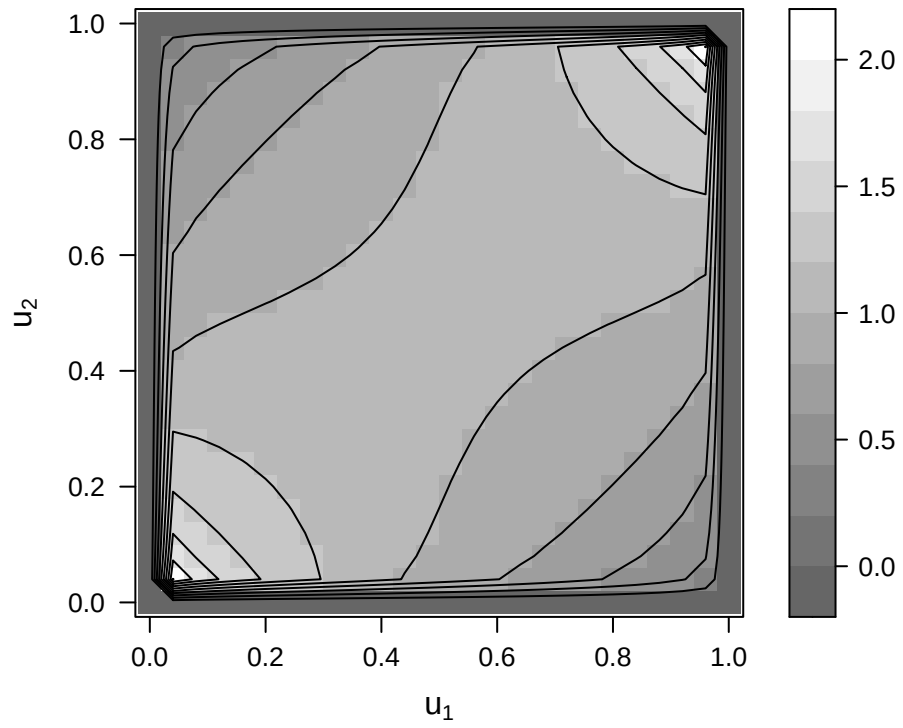
Theoretical frank Copula



Theoretical Clayton Copula



Theoretical Gumbel Copula



Theoretical Gumbel Copula

